

Clustered Federated Learning with Adaptive Similarity for Non-IID Data: Review Report

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Abstract

Federated Learning (FL) enables the collaborative training of models while preserving data privacy. However, statistical heterogeneity across non-IID data remains a major challenge that limits the performance of traditional approaches. This review examines the AS-CFL algorithm (Adaptive Similarity Clustered Federated Learning) proposed by Yi et al., which introduces dynamic clustering based on adaptive similarity together with a positive incentive mechanism. The approach uses low-rank matrix factorization to reduce complexity from $O(S^2d)$ to $O(Srd)$, and dynamically adjusts clusters through a dispersion-separation ratio. Results on MNIST and EMNIST show a 20% reduction in communication rounds while maintaining competitive accuracy. This review presents a critical analysis of the strengths, limitations, and future perspectives of this innovative approach.

Keywords: Federated learning, adaptive clustering, non-IID data, model similarity, privacy preservation.

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1 Introduction

Federated Learning (FL) has become a fundamental paradigm for distributed training of machine learning models, particularly in sensitive domains such as healthcare and finance, where data protection is critical [1]. By allowing devices to collaborate without sharing their raw data, FL meets the strict requirements of GDPR and HIPAA while still leveraging the collective power of distributed data.

1.1 The Challenge of Non-IID Data

The core problem in FL lies in the statistical heterogeneity of data, known as non-IID data (Non-Independent and Identically Distributed). This heterogeneity manifests as skewed label distributions, heterogeneous features, and quantitative imbalances between clients. These disparities lead to *client drift*, where local updates diverge from the global objective, resulting in slower convergence and reduced accuracy [2].

1.2 Clustering-Based Approaches and Their Limitations

Clustered Federated Learning (CFL) addresses this heterogeneity by grouping clients with similar distributions to train personalized per-cluster models [6]. However, existing methods present critical limitations:

- **Static approaches (IFCA [4]):** require the number of clusters to be predefined
- **Dynamic methods (FL+HC [5]):** prohibitive $O(N^2)$ complexity
- **No quality control:** no formal mechanism to evaluate client contribution

1.3 Contribution of AS-CFL

The AS-CFL algorithm proposed by Yi et al. [3] addresses these limitations through four major innovations: (1) adaptive dynamic clustering without a predefined K , (2) a positive incentive mechanism for quality control, (3) low-rank factorization for scalability, and (4) privacy preservation via differential noise. This review examines this approach, its results, and its future perspectives in depth.

2 AS-CFL Methodology

2.1 General Architecture

AS-CFL implements a cyclic three-phase workflow at every communication round (Figure 1), enabling continuous adaptation to evolving data distributions.

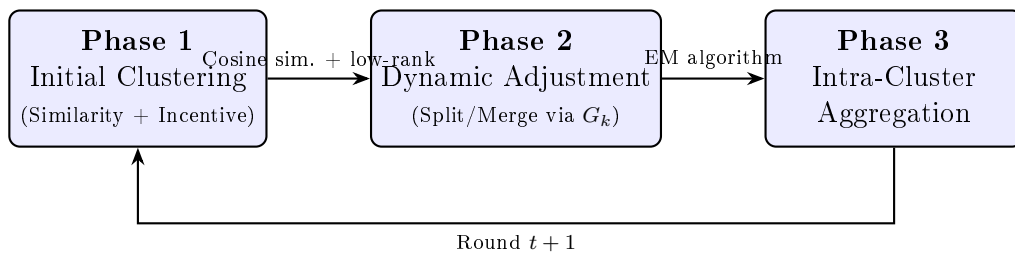


Figure 1: Three-phase workflow of the AS-CFL algorithm

2.2 Problem Formulation

For S clients with local datasets D_i , the goal is to partition the clients into K clusters, where each cluster k minimizes:

$$F_k(w_k) = \sum_{i \in S_k} \frac{|D_i|}{|D_k|} F_i(w_k) \quad (1)$$

The challenge lies in determining K and optimally assigning clients without direct access to their data distributions.

2.3 Phase 1: Adaptive Initial Clustering

2.3.1 Cosine Similarity and Low-Rank Approximation

AS-CFL measures similarity between clients through the cosine similarity of their model updates:

$$\text{sim}(i, j) = \frac{\Delta w_i \cdot \Delta w_j}{\|\Delta w_i\|_2 \|\Delta w_j\|_2} \quad (2)$$

To avoid the $O(S^2d)$ complexity of computing the full similarity matrix $M \in \mathbb{R}^{S \times S}$, AS-CFL applies a low-rank factorization:

$$M \approx U \Sigma V^T, \quad U, V \in \mathbb{R}^{S \times r}, \Sigma \in \mathbb{R}^{r \times r}, r \ll S \quad (3)$$

This factorization, applied to the output-layer weights, reduces the complexity to $O(Srd)$. The rank r is chosen to retain $\geq 95\%$ of explained variance, preserving discriminative information while ensuring scalability.

2.3.2 Positive Incentive Mechanism

The key innovation of AS-CFL is its incentive mechanism: a client i joins cluster C_k only if its inclusion improves performance on an IID proxy dataset D_{test} :

$$\text{eval}(C_k, \text{client}_i) = \text{Accuracy}(w_{C_k \cup \{i\}}^k, D_{\text{test}}) > \theta \quad (4)$$

where θ is an adjustable threshold (optimized at 0.85 in the experiments). This mechanism guarantees cluster quality by filtering out divergent clients. For computational efficiency, only the top 20% most similar clients are evaluated, with early stopping on the first failure to improve by 3%.

2.3.3 Adjustment for Imbalance

To handle imbalanced dataset sizes, AS-CFL adjusts the number of local epochs:

$$T_i = \max\left(\left\lceil \frac{BS}{|D_i|} \right\rceil, 10\right) \quad (5)$$

guaranteeing meaningful updates even for small datasets.

2.4 Phase 2: Dynamic Adjustment

2.4.1 EM Algorithm and Reassignment

AS-CFL uses an Expectation-Maximization algorithm to dynamically adjust clusters. The E-step reassigns clients:

$$z_{ik} = \arg \min_k \|\Delta w_i - w_k\|_2^2 \quad (6)$$

The M-step updates the centers:

$$w_k = \frac{\sum_{i \in S_k} z_{ik} \Delta w_i}{\sum_{i \in S_k} z_{ik}} \quad (7)$$

2.4.2 Dispersion-Separation Ratio

To assess structural stability, AS-CFL introduces the ratio G_k :

$$G_k = \frac{d_{in}(k)}{d_{out}(k)} = \frac{\frac{1}{|S_k|} \sum_{i \in S_k} \|\Delta w_i - w_k\|_2}{\min_{j \neq k} \|w_k - w_j\|_2} \quad (8)$$

The decision rule is: if $G_k > \beta = 0.8$ (high dispersion / low separation), the cluster is split; if $G_k < \alpha = 0.2$ (high cohesion / good separation), it is merged with a similar cluster. This makes K fully data-driven.

2.5 Phase 3: Aggregation with Privacy

Each cluster’s model is updated through weighted aggregation:

$$w_k^{t+1} = w_k^t + \sum_{i \in S_k} \frac{|D_i|}{|D_k|} (\Delta w_i^t + \mathcal{N}(0, \sigma^2 I)) \quad (9)$$

where the Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ ensures differential privacy with parameters $\epsilon = 1.0$, $\delta = 10^{-5}$.

2.6 Complexity Analysis

Client side: no additional overhead (identical to FedAvg).

Server side: low-rank factorization reduces the main bottleneck from $O(S^2 d)$ to $O(Srd)$. The other operations (incentive mechanism $O(S)$, G_k computation over K clusters) are highly efficient. The trade-off is favorable: manageable computational cost per round against a 20% reduction in total rounds.

3 Experimental Results

3.1 Configuration

Datasets: MNIST (10 classes, 60K images) and EMNIST (62 classes, 671K images), partitioned across 20 clients via Dirichlet distributions with $\alpha \in \{0.8, 0.6, 0.4\}$ (lower = more heterogeneous). Imbalance simulated with sizes ranging from 10% to 60%. Feature bias introduced via rotations ($0^\circ, 90^\circ, 180^\circ, 270^\circ$).

Baselines: FedAvg (single global model), CFL (fixed clustering), IFCA (alternating clustering/optimization).

Hyperparameters: learning rate 0.1 (decay 0.99), batch size 32, 10 local epochs.

3.2 Convergence and Accuracy

Figure 2 shows that AS-CFL reaches 90% accuracy in 40 rounds with a stable trajectory, whereas CFL requires 60 rounds. AS-CFL demonstrates faster and more reliable convergence thanks to its dynamic clustering.

Table 1 reveals that at low heterogeneity ($\alpha = 0.8$), AS-CFL and CFL are comparable. As heterogeneity increases, AS-CFL degrades more gracefully and outperforms CFL. On EMNIST at $\alpha = 0.4$, AS-CFL (0.682) significantly outperforms CFL (0.675) and FedAvg (0.500).

3.3 Communication Efficiency

AS-CFL achieves a 20% reduction in rounds compared to CFL. Low-rank aggregation reduces uplink traffic by 30% (28.3 MB $\rightarrow$ 19.8 MB). Compared to FedAvg, total communication cost drops by 40%.

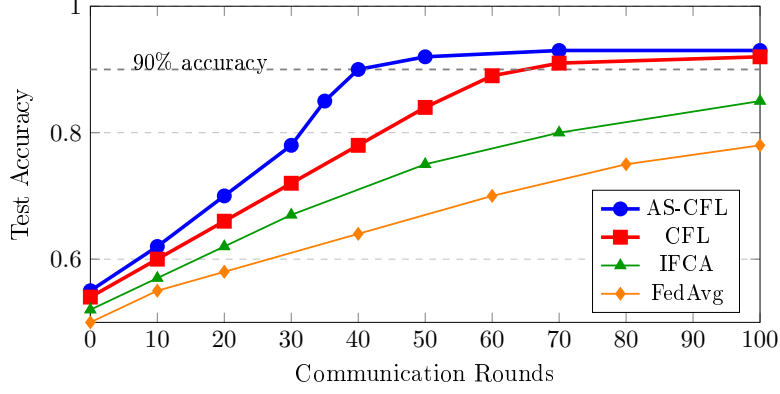


Figure 2: Convergence curves on EMNIST ($\alpha = 0.6$)

Table 1: Classification Accuracy Under Different Levels of Heterogeneity

Dataset	Algorithm	$\alpha=0.8$	$\alpha=0.6$	$\alpha=0.4$
MNIST	AS-CFL	0.917	0.915	0.910
	CFL	0.918	0.912	0.905
	IFCA	0.890	0.882	0.865
	FedAvg	0.875	0.850	0.805
EMNIST	AS-CFL	0.693	0.688	0.682
	CFL	0.695	0.685	0.675
	IFCA	0.620	0.605	0.580
	FedAvg	0.580	0.545	0.500

3.4 Robustness and Ablation

Robustness: with 10% random dropout per round, accuracy remains stable (0.905 vs. 0.908), demonstrating resilience thanks to dynamic EM reassignment.

Table 2: Ablation Study on MNIST ($\alpha = 0.6$)

Component	Accuracy	Rounds (90%)
Full AS-CFL	0.915	40
Without Positive Incentive	0.885 (-3.3%)	55 (+37.5%)
Without Dynamic Adjustment	0.902 (-1.4%)	48 (+20%)
Without Low-Rank	0.921 (+0.7%)	52 (+30%)

Table 2 shows that the incentive mechanism is the most critical component (3.3% accuracy loss, +37.5% rounds without it). Low-rank offers a marginal accuracy gain (+0.7%) but at the cost of 30% more rounds, justifying the accuracy-efficiency trade-off.

3.5 Sensitivity to Hyperparameters

AS-CFL degrades more gracefully than CFL as heterogeneity increases (decreasing α), with a growing performance gap in favor of AS-CFL under highly heterogeneous conditions. Cluster stability (Jaccard similarity between rounds) peaks at $\theta = 0.85$, confirming robustness within the 0.80–0.90 range.

4 Critical Analysis

4.1 Major Strengths

4.1.1 Methodological Innovations

AS-CFL brings four substantial contributions:

1. **Truly dynamic clustering**: eliminates the need to predefine K through the split/merge mechanism based on G_k , crucial for real-world applications where the population structure is unknown and evolving.
2. **Positive incentive mechanism**: formal evaluation of each client’s contribution, introducing quality control absent from purely distance-based methods. This innovation conceptually distinguishes AS-CFL.
3. **Algorithmic scalability**: reduction from $O(S^2d)$ to $O(Srd)$ enabling deployment on large client populations, often neglected in the CFL literature.
4. **Built-in privacy**: native incorporation of differentially private noise, aligning AS-CFL with data-protection regulations.

4.1.2 Experimental Validation

The experiments concretely demonstrate: a 20% reduction in communication rounds, superior robustness under strongly non-IID conditions, graceful degradation as heterogeneity increases, and substantial gains over FedAvg/IFCA.

4.2 Identified Limitations

4.2.1 Limited Experimental Datasets

The most critical limitation concerns the validation datasets. MNIST and EMNIST, although fundamental for comparison with reference CFL work, are relatively simple (28×28 grayscale pixels). The simulated non-IID patterns (Dirichlet, rotations) may not capture the complexity of natural heterogeneity found in real medical, financial, or IoT applications. The observed gains may not generalize to domains with distinct types of heterogeneity.

4.2.2 Open Questions About the Incentive Mechanism

Several questions remain:

- **Dependence on the proxy D_{test}** : in highly specialized domains (rare medical imaging), obtaining a representative IID dataset can be problematic or even impossible.
- **Computational cost**: despite pruning (top 20%), repeated forward passes for evaluation can become non-negligible for very large populations.
- **Empirical choice of θ** : although robust within 0.80–0.90, the choice of $\theta = 0.85$ remains empirical. An automatic adaptation mechanism could improve generality.

4.2.3 Thresholds α and β for G_k

The thresholds $\alpha = 0.2$ and $\beta = 0.8$ appear to be empirically determined without in-depth theoretical analysis. Their behavior on datasets with very different geometric characteristics (high dimensionality, multimodal distributions) remains to be explored. A risk of oscillation (repeated splits/merges) potentially exists.

4.2.4 Low-Rank Trade-off

The ablation study reveals an interesting point: the full matrix slightly improves accuracy (+0.6%) but increases rounds by 30%. A hybrid approach (low-rank initially, then full matrix toward the end of training) could offer the best of both worlds.

4.2.5 Missing Comparisons

The paper does not directly compare against certain recent adaptive CFL methods (FedAC, AdaCFL, FedLC) mentioned in its own literature review. A direct empirical comparison would have strengthened the validation.

4.3 Deployment Considerations

Impact of differential noise: the paper does not precisely quantify the accuracy degradation caused by noise, nor the additional communication overhead. Is the budget $\epsilon = 1.0$ sufficiently strict for sensitive applications?

Systemic heterogeneity: the paper focuses on statistical heterogeneity but does not deeply address variability in computational resources, variable network latency, or intermittent availability beyond the 10% dropout test.

4.4 Positioning in the State of the Art

Table 3: Qualitative Comparison with the State of the Art

Method	Adaptive K	Complexity	Quality Mechanism	Limitation
IFCA	No	$O(SK)$	None	Predefined K
FL+HC	Yes	$O(S^2d)$	None	Non-scalable
FedAC	Yes	$O(SK)$	None	Sensitive to K-Means
AS-CFL	Yes	$O(Srd)$	Incentive	Limited datasets

Table 3 shows that AS-CFL advantageously combines dynamic adaptability, computational scalability, and a unique quality-control mechanism.

4.5 Synthesis

AS-CFL represents a solid contribution to CFL, with well-designed innovations addressing real limitations. However, for AS-CFL to establish itself as a reference solution, future work will need to address: validation on complex/diverse datasets, theoretical convergence analysis, comparisons with recent methods, study of the privacy mechanism’s impact, and experiments in systemically heterogeneous environments.

5 Research Perspectives

5.1 Methodological Extensions

5.1.1 Modern Architectures

Transformers and foundation models: adapt low-rank factorization to multi-head attention mechanisms. Integration with PEFT (LoRA), where clustering would be performed on the low-rank adapters themselves.

Hierarchical clustering: extension toward multi-scale clustering enabling personalization at several levels (macro-cluster $\rightarrow$ micro-cluster $\rightarrow$ personal), better capturing complex structures.

5.1.2 Advanced Incentive Mechanisms

- **Bayesian approach:** estimate the probability that a client improves the cluster, incorporating uncertainty rather than a fixed threshold.
- **Reinforcement learning:** formulate selection as a multi-armed bandit problem where the server progressively learns which clients are beneficial.
- **Marginal contribution:** measure a client’s differential contribution relative to other cluster members.

5.2 Efficiency Optimization

Advanced compression: adaptive quantization based on parameter importance, structured sparsification, context-aware compression using cluster knowledge.

Parallelization: multi-server federated clustering, asynchronous aggregation for latency heterogeneity, GPU-acceleration of G_k computation.

5.3 Theoretical Guarantees

Convergence analysis: establish convergence bounds for AS-CFL under non-IID assumptions, analyze the impact of the split/merge mechanism on stability, determine optimal equilibrium conditions.

Formal privacy: rigorous proof of (ϵ, δ) -differential privacy for the entire AS-CFL process, cumulative budget analysis, theoretical privacy-utility trade-offs, extension to local privacy guarantees.

5.4 Emerging Applications

Digital health: personalized diagnosis by sub-population, federated drug discovery, privacy-preserving epidemiological surveillance.

Finance: personalized risk models, adaptive fraud detection, algorithmic trading preserving proprietary strategies.

IoT and Edge: predictive maintenance for Industry 4.0, smart cities with district-level models, autonomous vehicles adapted to regional conditions.

Continual learning: integration with lifelong learning for adaptation to concept drift, prevention of catastrophic forgetting, sequential multi-task learning.

5.5 Extended Validation

Complex vision datasets (CIFAR-10/100, ImageNet, medical datasets such as ISIC), non-image data (text, time series, graphs), standard FL benchmarks (LEAF, FLSim), natural heterogeneity (real multi-site medical data).

5.6 Ethical Considerations

Fairness: fairness-aware clustering avoiding clusters formed along sensitive demographic lines, equitable performance across clusters, representation of minority populations.

Interpretability: visualization and interpretation of cluster structures, explanations to clients regarding their assignment, auditability for regulatory compliance.

5.7 Unified Framework

Long-term vision: a unified adaptive framework combining dynamic clustering (AS-CFL), intelligent client selection (Active FL), cross-silo/cross-device optimization, federated AutoML, and multi-level privacy, automatically adapting to deployment characteristics.

6 Conclusion

6.1 Summary of Contributions

AS-CFL introduces four major innovations: truly dynamic clustering via the ratio G_k , a positive incentive mechanism for quality control, scalability optimization reducing complexity to $O(Srd)$, and native integration of differential privacy. These innovations address critical limitations of existing CFL approaches.

6.2 Main Results

Experiments on MNIST and EMNIST demonstrate a 20% reduction in communication rounds, competitive accuracy with growing superiority in strongly heterogeneous environments, robustness to dropout, and the necessity of each component validated through ablation.

6.3 Limitations and Perspectives

The identified limitations include: validation on simplistic datasets, absence of theoretical convergence analysis, empirical thresholds without in-depth justification, and missing comparisons with recent methods. Future perspectives cover methodological extensions (hierarchical clustering, advanced incentive mechanisms), efficiency optimizations, strengthened theoretical guarantees, emerging applications (healthcare, finance, IoT), and ethical considerations (fairness, interpretability).

6.4 Impact

Despite its limitations, AS-CFL represents a substantial advance toward practical, personalized federated learning systems. Its approach, balancing computational efficiency, communication efficiency, and clustering quality, establishes a new standard for CFL methods. The principles of AS-CFL – adaptability, contribution evaluation, and scalability – will provide a solid foundation for next-generation systems.

The transition to real-world deployments will require substantial additional work, particularly to establish theoretical guarantees and validate in genuinely heterogeneous environments. Nevertheless, AS-CFL constitutes a significant step toward achieving collaborative, personalized, and privacy-preserving machine learning at a global scale.

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